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Night vision goggle recording adapter

Abstract

A night vision goggle recording adapter includes a hollow body configured to receive a distal end of a diopter housing of a night vision goggle and a beam splitter disposed in a first interior space delimited by the hollow body. The beam splitter is configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body. Additional features include but are not limited to a diameter of the first interior space being smaller than an outer diameter of the diopter lens assembly, an image recording device adapter mounted in a removeable manner to the hollow body, and a removeable eyecup.

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Background/Summary

BACKGROUND

(1) Analog image intensifier tube technology enables users of night vision goggles (NVGs) to see in the dark. The analog nature does not lend toward recording, but digital NVG solutions do not match the sensitivity of analog NVGs. However, analog NVG images must be converted to digital for both recording and streaming of NVG images. Whether for training, promotion material, connecting NVG users digitally in real-time, etc., the demand for NVG recording solutions is growing. NVG recording solutions are constrained by metrics of space, weight, and power, as well as captured image quality, digital capabilities, and overall user experience. Thus, there is a need for an improved adapter for NVG recording.

SUMMARY

- (2) According to an embodiment of a night vision goggle recording adapter (NVGRA), the NVGRA comprises: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; and a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body, wherein a diameter of the first interior space is smaller than an outer diameter of the diopter lens assembly, such that a first wall of the hollow body that delimits the first interior space has an end face configured to abut a front face of the diopter lens assembly for all zero and positive diopter settings of the diopter lens assembly.
- (3) According to another embodiment of an NVGRA, the NVGRA comprises: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.
- (4) According to another embodiment of an NVGRA, the NVGRA comprises: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and a removeable eyecup slideably engaged onto the first wall of the hollow body that delimits the first interior space.
- (5) Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.
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Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The elements of the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts. The features of the various illustrated

embodiments can be combined unless they exclude each other. Embodiments are depicted in the drawings and are detailed in the description which follows.

(2) FIG. 1 illustrates a first side perspective view of a night vision goggle recording adapter (NVGRA) mounted to a diopter housing of a night vision goggles (NVG), according to an embodiment.

(3) FIG. 2 illustrates a second side perspective view of the NVGRA mounted to the diopter housing of the NVG.

(4) FIG. 3 illustrates a cross-sectional view of the NVGRA along the line labelled A-A' in FIGS. 1 and 2, in the horizontal X plane.

(5) FIG. 4 illustrates a backside perspective view of an image recording device adapter of the NVGRA.

(6) FIG. 5 illustrates a horizontal cross-sectional side perspective view of the NVGRA, according to another embodiment.

(7) FIG. 6 illustrates a top perspective view of the NVGRA and FIG. 7 illustrates a partial side perspective view of the NVGRA, according to another embodiment.

(8) FIG. 8 illustrates a horizontal cross-sectional top perspective view of the NVGRA, according to another embodiment.

(9) FIG. 9 illustrates a horizontal cross-sectional top perspective view of the NVGRA, according to another embodiment.

DETAILED DESCRIPTION

(10) Described herein are embodiments of a night vision goggle recording adapter (NVGRA). The NVGRA includes a hollow body configured to receive a distal end of a diopter housing of a night vision goggle and a beam splitter disposed in a first interior space delimited by the hollow body. The beam splitter is configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body. The NVGRA captures the entire night vision goggle (NVG) tube image without obstructing the user's view. The NVGRA is positioned within the confined space between an NVG user's face, their eye protection, helmet, and the diopter lens of the NVG. The NVGRA provides a full NVG image to any action camera by reflecting a fraction of the light exiting the NVG tube with the beam splitter.

(11) The NVGRA has various features which may be incorporated individually or in any combination, to address space, weight, and power constraints, and to improve captured image quality, digital capabilities, and overall user experience. Among these features are a diopter dip feature, a bent beam splitter feature, a removeable eyecup feature, and a removeable image recording device adapter feature.

(12) The diopter dip feature provides for a first wall of the hollow body that delimits the first interior space having an end face configured to abut a front face of the diopter lens assembly for all zero and positive diopter settings of the diopter lens assembly. The tailored geometry that dips into the NVG diopter lens assembly allows for full image capture and reduces how far the recorder extends towards the operator's face.

(13) The bent beam splitter feature provides for a first part of the beam splitter being angled with respect to the diopter lens assembly such that the beam of light that exits the diopter lens assembly toward the first part of the beam splitter has an angle of incidence less than 90 degrees. A second part of the beam splitter extends from the first part and is oriented at a different angle than the first part such that the first and second parts of the beam splitter collectively form a cover and the NVGRA is devoid of a separate cover lens at the viewing port. The bent beam splitter may be implemented using a polycarbonate material, for example.

(14) The removeable eyecup feature enhances useability of the NVGRA. Spatially, the user's nose extends more toward the NVG diopter than does the outside of his or her face. The removeable

eyecup feature exploits this to provide significant spatial footprint reductions. When installed on the NVGRA, the removeable eyecup feature reduces the amount of visible light that can pollute the recorded image. The beam splitter acts as a two-way mirror. In a dark room, only the NVG image is captured. When visible light is present, some image of the user's face will be present—generally outside of the NVG image projected onto the beam splitter. This may be more of an issue for training and promotional material than in an operational scenario, and thus the user can decide when to install the removeable eyecup.

(15) The removeable image recording device adapter feature allows for accommodating various types of image recording devices. Rather than create a unique NVGRA body for each camera in existence, individual image recording device adapters can be installed on the NVGRA. This allows one NVGRA unit to be compatible with more than one type of image recording device, or family of image recording devices. Additionally, future image recording devices can be readily adapted simply by producing a new image recording device adapter. Still other features and benefits of the NVGRA are described herein.

(16) Described next, with reference to the figures, are exemplary embodiments of the NVGRA.

(17) FIG. 1 illustrates a first side perspective view of the NVGRA **100** mounted to a diopter housing **102** of an NVG **104**. FIG. 2 illustrates a second side perspective view of the NVGRA **100** mounted to the diopter housing **102** of the NVG **104**. FIG. 3 illustrates a cross-sectional view of the NVGRA **100** along the line labelled A-A' in FIGS. 1 and 2, in the horizontal X plane. The term 'diopter' as used herein refers to the optical power of a lens or curved mirror and has a physical quantity equal to the reciprocal of the focal length, expressed in meters. For example, a 3-diopter lens brings parallel rays of light to focus at $\frac{1}{3}$ of a meter.

(18) The NVGRA **100** includes a hollow body **106** configured to receive a distal end **108** of the diopter housing **102** of the NVG **104**. The hollow body **106** may be formed by injection molding, machining, 3-D printing, etc.

(19) A beam splitter **110** is disposed in a first interior space **112** delimited by the hollow body **106** of the NVGRA **100**. The beam splitter **110** is configured to split a beam of light **114** that exits a diopter lens assembly **116** disposed in the diopter housing **102** of the NVG **104** into a transmitted beam **117** that exits the NVGRA **100** through a first (viewing) port **118** in the hollow body **106** of the NVGRA **100** and a reflected beam **120** that exits the NVGRA **100** through a second (recording) port **122** in the hollow body **100**. The diopter lens assembly **116** includes a lens **123** for focussing light captured by the diopter housing **102**.

(20) As shown in FIG. 3, a diameter D1 of the first interior space **112** of the hollow body **106** of the NVGRA **100** is smaller than an outer diameter D2 of the diopter lens assembly **116** of the NVG **104**. According to this embodiment, a first wall **124** of the hollow body **106** that delimits the first interior space **112** has an end face **126** configured to abut a front face **128** of the diopter lens assembly **116** for all zero and positive diopter settings of the diopter lens assembly **116** relative to the front face **130** of the diopter housing **102**, as opposed to a fixed justification such as +2 mm outside the diopter lens assembly **116**. The diameter relationship of $D1 < D2$ enables the diopter dip feature described above, and allows the user to select an optimal mounting position for the NVGRA **100** relative to the user's preferred diopter setting for the NVG **104** instead of a mounting position that is based on an arbitrary diopter setting not determined by the user (e.g., fixed at +2 diopters).

(21) A zero diopter setting means that the front face **128** of the diopter lens assembly **116** is flush with the front face **130** of the diopter housing **102**. Traversing the diopter lens assembly **116** outward from the zero setting correlates to a positive diopter adjustment. A positive diopter setting means that the front face **128** of the diopter lens assembly **116** is extended outward beyond the front face **130** of the diopter housing **102**. A negative diopter setting means that the front face **128** of the diopter lens assembly **116** is recessed inward from the front face **130** of the diopter housing **102**.

(22) The diopter dip feature can account for diopter adjustments from -1 to +4 or even more

positive, e.g., +6 diopters. This translates to space savings between the NVGRA **100** and the user's face, as the beam splitter **110** still captures the image regardless of diopter adjustment while also retaining maximal image alignment while traversing with the diopter housing **102**.

(23) Many NVGs have an adjustable diopter setting, e.g., in a range of -6 to +6 diopters, which represents approximately 5 mm travel in the Y axis in FIG. 3. As shown in FIG. 3, the beam splitter geometry may enable the beam splitter **110** to extend into the diopter lens assembly **116** when the first wall **124** of the hollow body **106** that delimits the first interior space **112** of the NVGRA **100** abuts the front face **128** of the diopter lens assembly **116**. This allows the beam splitter **110** to be positioned some distance inside of the diopter lens assembly **116** in the Y axis.

(24) To reduce interference with the diopter housing **102**, the outer edge **132** of the beam splitter **110** that faces toward the diopter lens assembly **116** may be bevelled, allowing a more precise fitment while retaining enough of the beam splitter **110** to capture the inner edge of the NVG image **114**. If the diameter D1 of the first interior space **112** of the hollow body **106** of the NVGRA **100** is smaller than the outer diameter D2 of the diopter lens assembly **116** of the NVG **104**, the beam splitter **110** is justified at least through the full range of zero and all positive diopter settings by riding the body of the diopter lens assembly **116** rather than the outer face **134** of the diopter housing **102** and enabling full traversal for adjustable diopter assemblies.

(25) As shown in FIG. 3, the hollow body **106** of the NVGRA **100** may also have a second wall **136** that delimits a second interior space **138** configured to receive the distal end **108** of the diopter housing **102**. The first interior space **112** of the hollow body **106** of the NVGRA **100** is openly connected to the second interior space **138**. The first wall **124** and the second wall **136** of the hollow body **106** of the NVGRA **100** are connected to one another by a transition region **140** of the hollow body **106** that accommodates an outer diameter difference (D3-D2) between the diopter housing **102** and the diopter lens assembly **116** of the NVG **104**. As shown in FIGS. 1 and 2, the NVGRA **100** may include a clamp or similar type of fixing structure **142** for holding the NVGRA **100** in place when mounted to the diopter housing **102** of the NVG **104**.

(26) FIG. 3 also illustrates the removeable image recording device adapter feature described above. In FIG. 3, the NVGRA **100** may also include an image recording device adapter **144** mounted in a removeable manner to the hollow body **106** of the NVGRA **100** and aligned with the second (recording) port **122** in the hollow body **106**. The image recording device adapter **144** and the hollow body **106** are separate parts in this embodiment. The image recording device adapter **144** is configured to receive an image recording device **146** such as an action camera and pass the beam **120** reflected by the beam splitter **110** through the second (recording) port **122** of the NVGRA **100** to the image recording device **146** via a corresponding port **148** in the image recording device adapter **144**.

(27) FIG. 2 shows an image recording device **146** mounted to the image recording device adapter **144** of the NVGRA **100**. The image recording device **146** may be secured to the image recording device adapter **144** by a vinyl strap (not shown) that provides a gripping surface and positive retention, for example. The image recording device adapter **144** may be formed by injection molding, machining, 3-D printing, etc.

(28) FIG. 4 illustrates a backside perspective view of the image recording device adapter **144**. In FIG. 4, the backside of the image recording device adapter **144** has one or more surface features **150** such as ridges, walls, lips, etc., that mate with the second (recording) port **122** in the hollow body **106** of the NVGRA **100**. One or more openings **152** may be formed in the image recording device adapter **144** and through which a fastener (not shown) such as a screw can be inserted and used to secure the image recording device adapter **144** to the hollow body **106** of the NVGRA **100**. To accommodate a different type of image recording device **146**, the image recording device adapter **144** is removed from the hollow body **106** of the NVGRA **100** and replaced with an adapter designed for that device.

(29) FIG. 5, which is a horizontal cross-sectional side perspective view of the NVGRA **100**,

illustrates the bent beam splitter feature described above. In FIG. 5, the beam splitter **110** includes a first part **154** and a second part **156**. The first part **154** is angled with respect to the diopter lens assembly **116** such that the beam of light **114** that exits the diopter lens assembly **116** toward the first part **154** of the beam splitter **110** has an angle α_1 of incidence less than 90 degrees. The second part **156** of the beam splitter **110** extends from the first part **154** and is oriented at a different angle α_2 than the first part **154** of the beam splitter **110**. The first and second parts **154**, **156** of the beam splitter **110** collectively form a cover for the first (viewing) port **118** in the hollow body **106** of the NVGRA **100** such that the NVGRA **100** is devoid of a separate cover lens at the first (viewing) port **118** which saves about 3 mm, improving user comfort. In one embodiment, the beam splitter **110** is formed of a polycarbonate material to enable the bending illustrated in FIG. and 5 where the first and second parts **154**, **156** of the beam splitter **110** are formed as a single, contiguous unit.

(30) The image recording device adapter **144** is shown as an integral, contiguous part of the hollow body **106** of the NVGRA **100** in FIG. 5. According to this embodiment, the image recording device adapter **144** is not removeable. The image recording device adapter **144** instead may be implemented as shown in FIGS. 3 and 4, such that the image recording device adapter **144** is removeable.

(31) FIG. 6, which is a top perspective view of the NVGRA **100**, and FIG. 7, which is a partial side perspective view of the NVGRA **100**, illustrate the removeable eyecup feature described above. In FIGS. 6 and 7, the NVGRA **100** further includes a removeable eyecup **158** configured to slideably engage onto the first wall **124** of the hollow body **106** that delimits the first interior space **112** of the NVGRA **100**. FIG. 6 shows the removeable eyecup **158** in a demounted state, whereas FIG. 7 shows the removeable eyecup **158** in a mounted state.

(32) In one embodiment, the removeable eyecup **158** and the first wall **124** of the hollow body **106** that delimits the first interior space **112** of the NVGRA **100** are keyed in a complementary manner to yield mating surface structures **160**, **162** that enable the removeable eyecup **158** to slideably engage the first wall **124** of the hollow body **106** that delimits the first interior space **112** of the NVGRA **100**. For example, the mating surface structure **160** of the removeable eyecup **158** may include a slot or groove and the corresponding mating surface structure **162** of the first wall **124** of the hollow body **106** may include a ridge, tab, or other type of protrusion that is received by the slot/groove when the removeable eyecup **158** is slideably engaged onto the first wall **124** of the hollow body **106**.

(33) The removeable eyecup **158** may comprise a softer material such as a rubber material than the hollow body **106** of the NVGRA **100**, such that the removeable eyecup **158** yields to physical pressure more readily than the hollow body **106** to enhance user comfort when the removeable eyecup **158** is mounted to the hollow body **106**. The removeable eyecup **158** may break away when excessive force is applied, reducing user discomfort and preventing damage to the NVGRA **100**. The removeable eyecup **158** allows the user to choose a balance between intimacy with his or her face via closer positioning, or superior image insulation from exterior white light. The removeable eyecup **158** may be adapted for operators wanting no light leakage from the NVGRA **100** to the surrounding environment.

(34) FIG. 8 illustrates a horizontal cross-sectional top perspective view of the NVGRA **100**, according to another embodiment. In FIG. 8, the NVGRA **100** further includes an aperture stop and/or a corrective lens **164** placed in the viewing optical path. For example, the aperture stop and/or corrective lens **164** may be housed in the first (viewing) port **118** in the hollow body **106** of the NVGRA **100** as shown in FIG. 8.

(35) Some image recording devices such as action cameras have a wide aperture lens (e.g., ~F2.5), resulting in a shallow depth of field. This can result in a slightly out of focus image when recording NVG footage, with or without the beam splitter **110** reflecting the image at 90 degrees. To correct for this, the aperture stop and/or corrective lens **164** can be inserted in the viewing optical path, e.g., seated in the first (viewing) port **118** in the hollow body **106** of the NVGRA **100**. An aperture

stop limits the amount of light reaching the first (viewing) port **118** in the hollow body **106**, whereas a corrective lens has a corrective power that improves visual perception.

(36) FIG. **9** illustrates a horizontal cross-sectional top perspective view of the NVGRA **100**, according to another embodiment. In FIG. **9**, the NVGRA **100** has the removeable eyecup **158** and the aperture stop and/or corrective lens **164** is placed in the recording optical path instead of the viewing optical path. For example, as shown in FIG. **9**, the aperture stop and/or corrective lens **164** may be housed in the second (recording) port **122** in the hollow body **100** of the NVGRA **100**. In another example, the aperture stop and/or corrective lens **164** may be housed in the recording port **148** in the image recording device adapter **144**. In another embodiment, a first aperture stop and/or a first corrective lens **164** may be housed in the first (viewing) port **118** in the hollow body **106** of the NVGRA **100**, e.g., as shown in FIG. **8**, and a second aperture stop and/or a second corrective lens **164** may be housed in the second (recording) port **122** in the hollow body **100** of the NVGRA **100** or in the recording port **148** in the image recording device adapter **144**, e.g., as shown in FIG. **9**.

(37) The image recording device adapter **144** and the hollow body **106** of the NVGRA **100** are shown as separate parts in FIGS. **8** and **9**. According to these embodiments, the image recording device adapter **144** is removeable and the first wall **124** of the hollow body **106** of the NVGRA **100** has one or more openings **166** for receiving a fastener used to secure the removeable image recording device adapter **144** to the NVGRA **100**. The image recording device adapter **144** instead may be an integral, contiguous part of the hollow body **106** of the NVGRA **100**, e.g., as shown in FIG. **5** and therefore not removeable.

(38) Although the present disclosure is not so limited, the following numbered examples demonstrate one or more aspects of the disclosure.

(39) Example 1. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; and a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body, wherein a diameter of the first interior space is smaller than an outer diameter of the diopter lens assembly, such that a first wall of the hollow body that delimits the first interior space has an end face configured to abut a front face of the diopter lens assembly for all zero and positive diopter settings of the diopter lens assembly.

(40) Example 2. The night vision goggle recording adapter of example 1, wherein the beam splitter extends into the diopter lens assembly when the first wall of the hollow body that delimits the first interior space abuts the front face of the diopter lens assembly.

(41) Example 3. The night vision goggle recording adapter of example 1 or 2, wherein the beam splitter comprises a first part and a second part, wherein the first part is angled with respect to the diopter lens assembly such that the beam of light that exits the diopter lens assembly toward the first part of the beam splitter has an angle of incidence less than 90 degrees, wherein the second part extends from the first part and is oriented at a different angle than the first part, and wherein the first part and the second part of the beam splitter collectively form a cover for the first port in the hollow body such that the night vision goggle recording adapter is devoid of a separate cover lens at the first port.

(42) Example 4. The night vision goggle recording adapter of example 3, wherein the beam splitter is formed of a polycarbonate material.

(43) Example 5. The night vision goggle recording adapter of any of examples 1 through 4, further comprising: a removeable eyecup slideably engaged onto the first wall of the hollow body that delimits the first interior space.

(44) Example 6. The night vision goggle recording adapter of example 5, wherein the removeable

eyecup and the first wall of the hollow body that delimits the first interior space are keyed in a complementary manner that enables the removeable eyecup to slideably engage the first wall of the hollow body that delimits the first interior space.

(45) Example 7. The night vision goggle recording adapter of example 5 or 6, wherein the removeable eyecup comprises a softer material than the hollow body such that the removeable eyecup yields to physical pressure more readily than the hollow body.

(46) Example 8. The night vision goggle recording adapter of any of examples 1 through 7, further comprising: an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.

(47) Example 9. The night vision goggle recording adapter of example 8, further comprising: an aperture stop and/or a corrective lens housed in the image recording device adapter.

(48) Example 10. The night vision goggle recording adapter of any of examples 1 through 8, further comprising: an aperture stop and/or a corrective lens housed in the first port.

(49) Example 11. The night vision goggle recording adapter of any of examples 1 through 10, wherein the hollow body further comprises a second wall that delimits a second interior space configured to receive the distal end of the diopter housing, wherein the first interior space is openly connected to the second interior space, and wherein the first wall and the second wall are connected to one another by a transition region of the hollow body that accommodates an outer diameter difference between the diopter housing and the diopter lens assembly of the night vision goggle.

(50) Example 12. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.

(51) Example 13. The night vision goggle recording adapter of example 12, further comprising: an aperture stop and/or a corrective lens housed in the image recording device adapter.

(52) Example 14. The night vision goggle recording adapter of example 12 or 13, wherein the beam splitter comprises a first part and a second part, wherein the first part is angled with respect to the diopter lens assembly such that the beam of light that exits the diopter lens assembly toward the first part of the beam splitter has an angle of incidence less than 90 degrees, wherein the second part extends from the first part and is oriented at a different angle than the first part, and wherein the first part and the second part of the beam splitter collectively form a cover for the first port in the hollow body such that the night vision goggle recording adapter is devoid of a separate cover lens at the first port.

(53) Example 15. The night vision goggle recording adapter of example 14, wherein the beam splitter is formed of a polycarbonate material.

(54) Example 16. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and a removeable eyecup slideably engaged onto the first wall of the hollow body that delimits the

first interior space.

(55) Example 17. The night vision goggle recording adapter of example 16, wherein the removeable eyecup and the first wall of the hollow body that delimits the first interior space are keyed in a complementary manner that enables the removeable eyecup to slideably engage the first wall of the hollow body.

(56) Example 18. The night vision goggle recording adapter of example 16 or 17, wherein the removeable eyecup comprises a softer material than the hollow body such that the removeable eyecup yields to physical pressure more readily than the hollow body.

(57) Example 19. The night vision goggle recording adapter of any of examples 16 through 18, further comprising: an aperture stop and/or a corrective lens housed in the first port.

(58) Example 20. The night vision goggle recording adapter of any of examples 16 through 19, further comprising: an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.

(59) Terms such as “first”, “second”, and the like, are used to describe various elements, regions, sections, etc. and are also not intended to be limiting. Like terms refer to like elements throughout the description.

(60) As used herein, the terms “having”, “containing”, “including”, “comprising” and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles “a”, “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

(61) The expression “and/or” should be interpreted to include all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression “A and/or B” should be interpreted to mean only A, only B, or both A and B. The expression “at least one of” should be interpreted in the same manner as “and/or”, unless expressly noted otherwise. For example, the expression “at least one of A and B” should be interpreted to mean only A, only B, or both A and B.

(62) It is to be understood that the features of the various embodiments described herein can be combined with each other, unless specifically noted otherwise.

(63) Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; and a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body, wherein a diameter of the first interior space is smaller than an outer diameter of the diopter lens assembly, such that a first wall of the hollow body that delimits the first interior space has an end face configured to abut a front face of the diopter lens assembly for all zero and positive diopter settings of the diopter lens assembly.

2. The night vision goggle recording adapter of claim 1, wherein the beam splitter extends into the

diopter lens assembly when the first wall of the hollow body that delimits the first interior space abuts the front face of the diopter lens assembly.

3. The night vision goggle recording adapter of claim 1, wherein the beam splitter comprises a first part and a second part, wherein the first part is angled with respect to the diopter lens assembly such that the beam of light that exits the diopter lens assembly toward the first part of the beam splitter has an angle of incidence less than 90 degrees, wherein the second part extends from the first part and is oriented at a different angle than the first part, and wherein the first part and the second part of the beam splitter collectively form a cover for the first port in the hollow body such that the night vision goggle recording adapter is devoid of a separate cover lens at the first port.

4. The night vision goggle recording adapter of claim 3, wherein the beam splitter is formed of a polycarbonate material.

5. The night vision goggle recording adapter of claim 1, further comprising: a removeable eyecup slideably engaged onto the first wall of the hollow body that delimits the first interior space.

6. The night vision goggle recording adapter of claim 5, wherein the removeable eyecup and the first wall of the hollow body that delimits the first interior space are keyed in a complementary manner that enables the removeable eyecup to slideably engage the first wall of the hollow body that delimits the first interior space.

7. The night vision goggle recording adapter of claim 5, wherein the removeable eyecup comprises a softer material than the hollow body such that the removeable eyecup yields to physical pressure more readily than the hollow body.

8. The night vision goggle recording adapter of claim 1, further comprising: an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.

9. The night vision goggle recording adapter of claim 8, further comprising: an aperture stop and/or a corrective lens housed in the image recording device adapter.

10. The night vision goggle recording adapter of claim 1, further comprising: an aperture stop and/or a corrective lens housed in the first port.

11. The night vision goggle recording adapter of claim 1, wherein the hollow body further comprises a second wall that delimits a second interior space configured to receive the distal end of the diopter housing, wherein the first interior space is openly connected to the second interior space, and wherein the first wall and the second wall are connected to one another by a transition region of the hollow body that accommodates an outer diameter difference between the diopter housing and the diopter lens assembly of the night vision goggle.

12. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.

13. The night vision goggle recording adapter of claim 12, further comprising: an aperture stop and/or a corrective lens housed in the image recording device adapter.

14. The night vision goggle recording adapter of claim 12, wherein the beam splitter comprises a first part and a second part, wherein the first part is angled with respect to the diopter lens assembly such that the beam of light that exits the diopter lens assembly toward the first part of the beam

splitter has an angle of incidence less than 90 degrees, wherein the second part extends from the first part and is oriented at a different angle than the first part, and wherein the first part and the second part of the beam splitter collectively form a cover for the first port in the hollow body such that the night vision goggle recording adapter is devoid of a separate cover lens at the first port.

15. The night vision goggle recording adapter of claim 14, wherein the beam splitter is formed of a polycarbonate material.

16. A night vision goggle recording adapter, comprising: a hollow body configured to receive a distal end of a diopter housing of a night vision goggle; a beam splitter disposed in a first interior space delimited by the hollow body, the beam splitter configured to split a beam of light that exits a diopter lens assembly disposed in the diopter housing into a transmitted beam that exits the night vision goggle recording adapter through a first port in the hollow body and a reflected beam that exits the night vision goggle recording adapter through a second port in the hollow body; and a removeable eyecup slideably engaged onto a first wall of the hollow body that delimits the first interior space.

17. The night vision goggle recording adapter of claim 16, wherein the removeable eyecup and the first wall of the hollow body that delimits the first interior space are keyed in a complementary manner that enables the removeable eyecup to slideably engage the first wall of the hollow body.

18. The night vision goggle recording adapter of claim 16, wherein the removeable eyecup comprises a softer material than the hollow body such that the removeable eyecup yields to physical pressure more readily than the hollow body.

19. The night vision goggle recording adapter of claim 16, further comprising: an aperture stop and/or a corrective lens housed in the first port.

20. The night vision goggle recording adapter of claim 16, further comprising: an image recording device adapter mounted in a removeable manner to the hollow body and aligned with the second port in the hollow body, wherein the image recording device adapter is configured to receive an image recording device and pass the beam reflected by the beam splitter through the second port to the image recording device.
